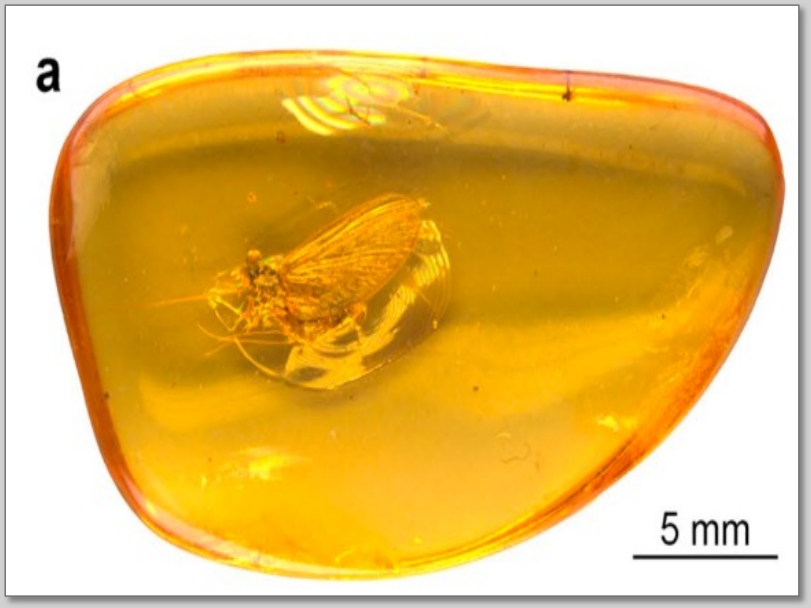
Nottingham Trent University

Quantitative Micro-CT Analysis of Insects in Resin

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Date: September 29th, 2026.

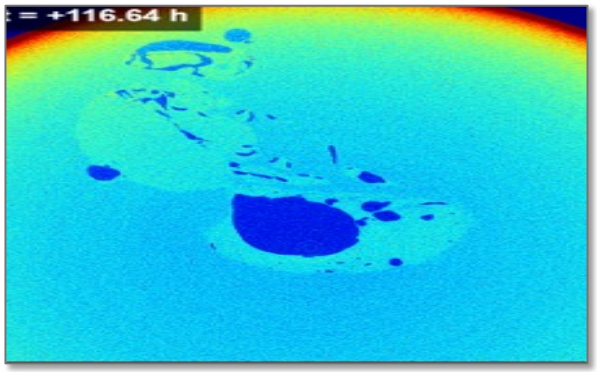


Insect preserved in natural fossil amber (Calliarcys antiquus sp. nov., Godunko et al 2022.,)

The Problem: Linear Attenuation Overlap in Polymer Matrices



Optical surface blocked by matrix



3D penetration but overlapping tissue-resin density

- **Context & Utility:**
  - **Specimen Preservation:** Synthetic resin enables safe handling and museum display.
  - **Natural Analogue:** Fossil amber preserves biological inclusions over geological timescales.
- **The Optical Bottleneck:**
  - Darkened or oxidized matrix boundaries block traditional light microscopy.
- **The Tomographic Segmentation Bottleneck:**
  - Micro-CT penetrates non-destructively, but automated segmentation fails with a single threshold fails.
  - Resin and biological tissue produce overlapping peaks in attenuation histograms.
- **Motivation:**
  - Develop unsupervised segmentation and multi-energy spectral PCA to isolate soft anatomy without stains.
  - Quantify embedding taphonomy and curing kinetics in resin-preserved specimens.

Research Aims & Objectives

OBJECTIVE 01: AUTOMATED SEGMENTATION

- **Unsupervised Computational Framework**
- **Spatial masking + dynamic KDE mode suppression pipeline**

OBJECTIVE 02:SPECTRAL PCA

- **Overcome low soft-tissue contrast ($\Delta\mu = 0$) without chemical stains**
- **Isolate correlated anatomical attenuation**

OBJECTIVE 03: TAPHONOMIC COMPARISON

- **Benchmark internal structural contrast at standardized 90 KV**
- **Compare fresh/hydrated vs. 6-month desiccated tissue preservation**

OBJECTIVE 04: IN-SITU KINETICS

- **Time-resolved micro-CT across embedding and polymer cross-linking**
- **Quantify moisture leaching, cavity gas expansion, and resin curing**

Experimental Methodology & Specimens

INVESTIGATED SPECIMENS

- **Amber Fossil**
- **Synthetic resin controls:**
- Hydrated Model: Fresh bee embedded directly (wet entombment).
- Dessicated Model: 6-month pre-dried bee (void & boundary formation).
- **Dynamic In-Situ Cohort:** Real-time CT monitoring during resin curing (0 - 27 h) and (17– 117 h)

01

Micro-CT Acquisition

- Multi-energy stacks (40 – 120 k) + standardized 90 kV scans.
- Time-series scan at 50 kV.

02

Pre-Processing

- Calibration with air reference points $I_{cal} = I_{raw} - (I_{air,current} - I_{air,ref})$

- Automated circular spatial masking to eliminate sample holders.

03

KDE Segmentation

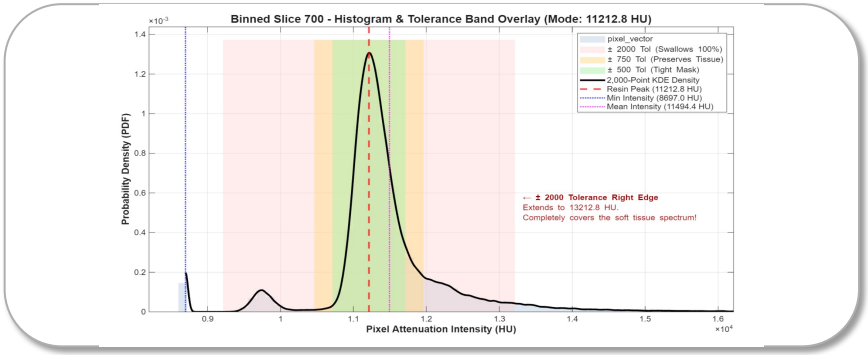
- Dynamic slice-by-slice mode suppression of bulk matrix peaks..

04

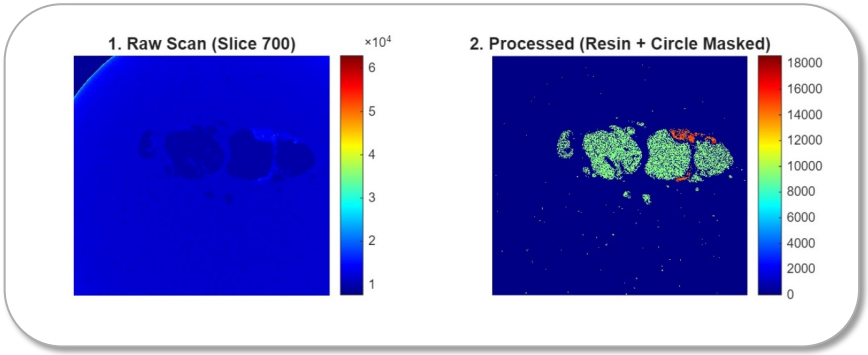
Spectral PCA

- Multi-kV covariance mapping to isolate biological anatomy.

Automated Segmentation: Spatial Masking & Dynamic KDE



Dynamic identification and thresholding of the dominant matrix peak per slice.



Left: Raw CT slice with resin background. Right: Automated KDE cuticle extraction.

Spatial Circular Masking 1

- Isolates region of interest; eliminates container boundary artifacts.
- Automated radial cutoff discards air and mold walls prior to density fitting.

Continuous Density Modeling (KDE) 2

- Overcomes discrete histogram binning artifacts and photon noise.
- Computes dynamic probability density curves per slice using Gaussian kernel smoothing.

Dynamic Mode Suppression 3

- Unsupervised background subtraction across variable axial geometries.
- Automatically suppresses dominant resin peak while preserving insect cuticle boundaries.

Multi-Energy Spectral PCA

Hydrated Soft-Tissue Invisibility 1
- Bulk resin and hydrated tissue share nearly identical attenuation ($\Delta\mu = 0$).
- Single-energy scans (90 – 120 kV) isolate only 0.64% - 1.02% of tissue.

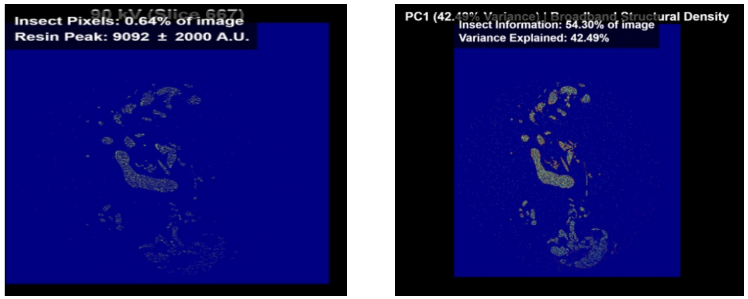
Cross-Channel Covariance Mapping 2
- Evaluates 40 – 120 kV acquisitions collectively as a multi-spectral stack.
- Leverages spatial correlation: real tissue attenuates across all channels; noise fluctuates randomly.

Signal Consolidation 3
- Covariance diagonalization concentrates maximum shared variance into PC_1.

BIOLOGICAL TISSUE CONTRAST ENHANCEMENT **60.9X**

Quantitative Information Gain in PC_1

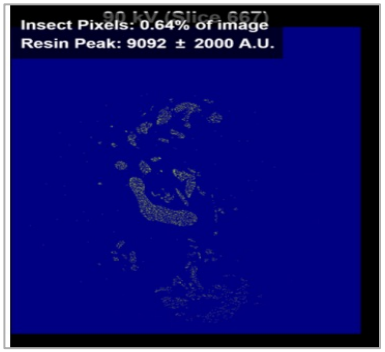
Single Energy (90 – 120 kV): 0.64% - 1.02% → Multi – Energy PC_1: 54.30%



Left: Segmented single energy image. Right: High-contrast tissue isolation in PC_1.

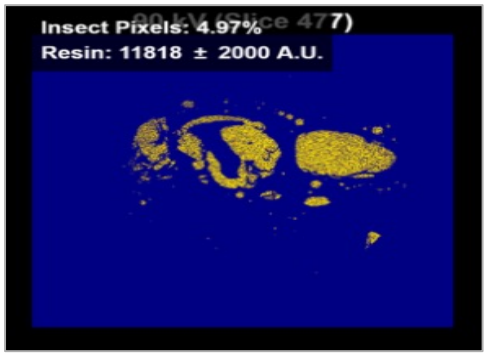
Taphonomic State Comparison: Hydrated vs. Desiccated

MODEL A: FRESH / HYDRATED

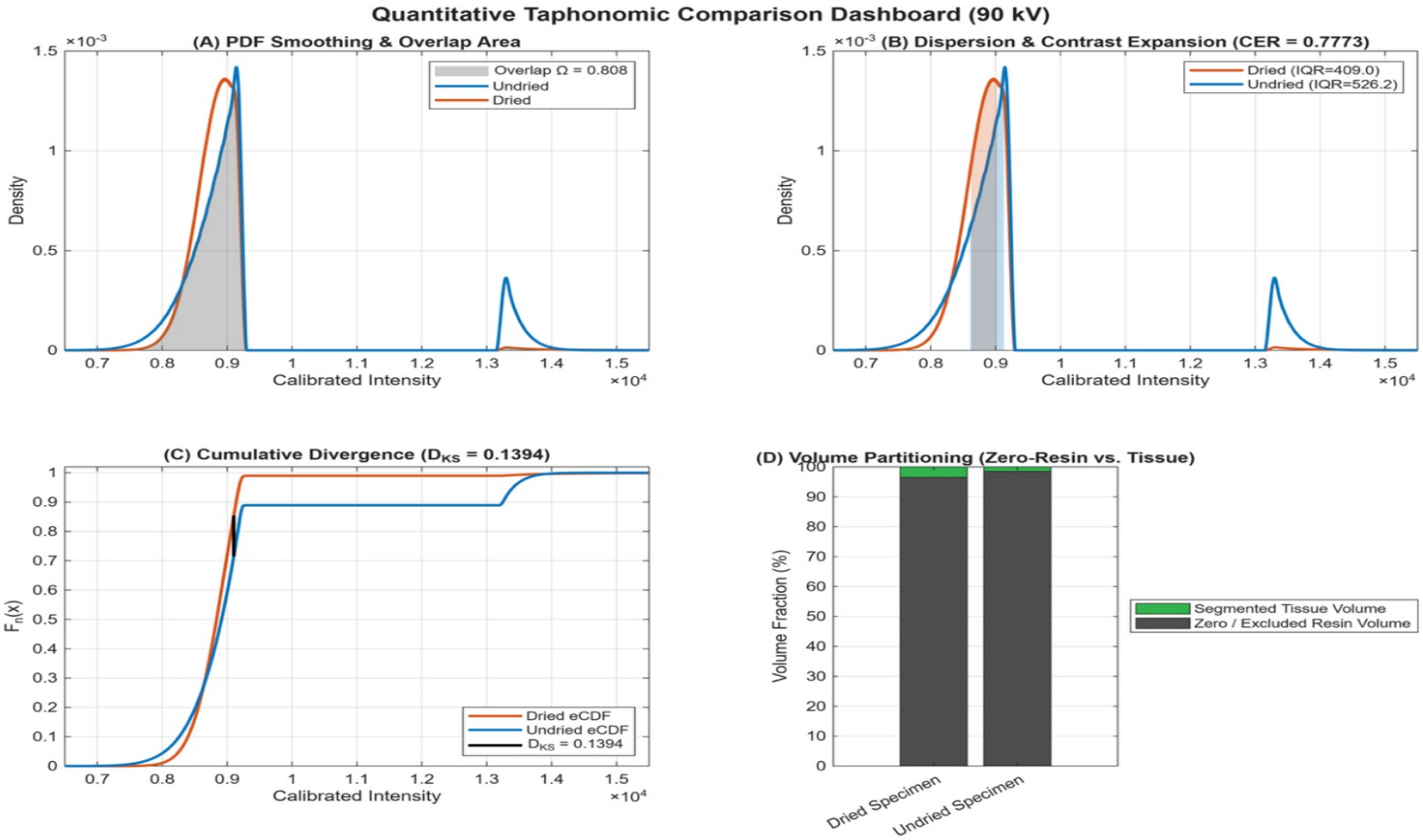


- **Density Overlap:** Tissue fluids remain trapped, causing soft tissues to match epoxy resin attenuation ($\Delta\mu = 0$)
- **Boundary Blurring:** Lack of density boundaries leads to complete soft-tissue erasure during conventional thresholding.
- **Fossil Analogue:** Reflects live or immediate resin engulfment prior to cellular dehydration.

MODEL B: 6-MONTH DESICCATED (PRE-DRIED)

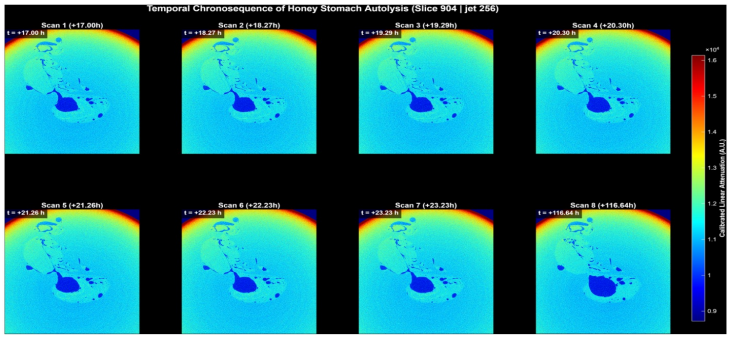


- **Shrinkage & Cavities:** Moisture loss causes viscera to shrink, forming air/gas pockets within the exoskeleton**.**
- **Natural Negative Contrast:** The near-zero attenuation of internal air voids provides sharp structural differentiation without chemical stains**.**
- **Fossil Analogue:** Mirrors prolonged surface exposure and partial decay prior to resin entombment.



Quantitative comparison of the concatenated pixel intensities for the dried and undried bees.

In-Situ Curing Kinetics & Dynamic Dehydration



Sequential cross-sections illustrating progressive internal void enlargement and cuticular detachment over 117 hours.

TAPHONOMIC IMPLICATION

Biological decay and negative-contrast cavity formation begin immediately upon entombment, proving that internal air pockets in fossil amber can originate from early-stage syn-depositional desiccation rather than post-fossilization degradation.

Moisture Leaching & Boundary Diffusion 1

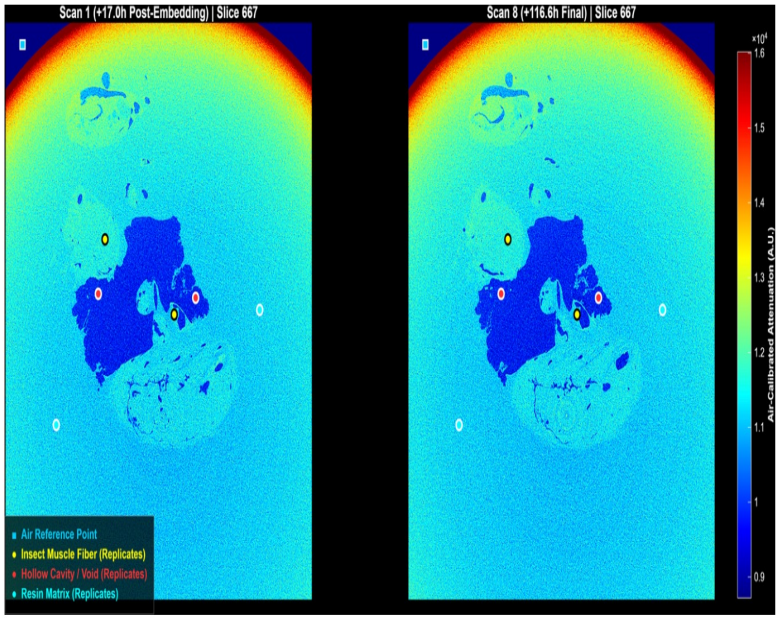
- Dynamic water exudation across cuticular interfaces during initial liquid resin contact.
- Rapid establishment of a localized diffusion gradient at specimen boundaries.

Internal Cavitation & Tissue Collapse 2

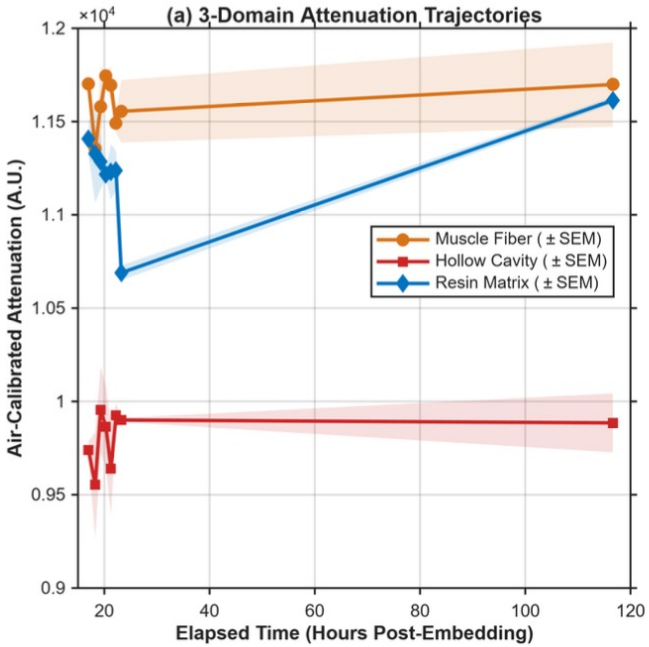
- Progressive visceral dehydration leads to significant internal volume shrinkage.
- Internal gas voids expand monotonically, generating high-contrast negative attenuation boundaries.

Matrix Densification & Cross-Linking 3

- Exothermic cross-linking drives volumetric contraction and uniform matrix densification.
- Distinguishes mechanical resin shrinkage from genuine biological degradation.



Spatial sampling locations on Slice 667 demonstrate the high-density thoracic muscle bundles relative to internal hollow regions at both early (+17.00 h) and final (+116.64 h) stages.



Three-domain attenuation curves across the 117-hour timeframe confirmed long-term thoracic muscle stability alongside progressive resin matrix hardening

Key Findings & Scientific Contributions

COMPUTATIONAL ADVANCEMENT

Dynamic KDE Mode Suppression

- Automated circular spatial masking successfully isolates sample volume from container artifacts.
- Continuous Gaussian density modeling resolves the > 90% resin volume dominance without manual user bias.
- Generalizes across both synthetic resin matrices and dark, oxidized Baltic amber inclusions.

QUANTITATIVE RECOVERY

Multi-Energy PCA (PC_1)

- Resolves the zero-attenuation boundary problem ($\Delta\mu = 0$) in hydrated tissues without destructive chemical stains.
- Delivers a 60.95 X quantitative information gain in internal soft-tissue recovery over single-energy CT.
- Decorrelates true biological signal from random photon noise and detector ring artifacts.

PALEOBIOLOGICAL IMPACT

Syn-Depositional Kinetics

- Establishes that pre-entombment desiccation generates natural negative contrast via internal air/gas voids.
- In-situ time-series CT confirms visceral dehydration begins immediately upon resin contact (0 – 27 h)
- Demonstrates internal void morphology in fossils reflects early resin entrapment rather than post-burial decay.

Current Limitations & Future Directions

METHODOLOGICAL CONSTRAINTS

Physical & Computational Boundaries

- Polychromatic Beam Hardening: Standard laboratory benchtop sources produce cupping and streak artifacts that skew subtle tissue-matrix density distinctions.
- Spatial Resolution Trade-Offs: Whole-specimen fields of view constrain voxel resolution, limiting detection of subcellular cuticular ultrastructure.
- Computational Overhead: Multi-energy spectral stacks require extensive slice-by-slice co-registration and covariance matrix calculations.

HIGH-IMPACT EXTENSIONS

Advanced Modalities & Deep Learning

- Synchrotron Radiation (SR-CT): Harness monochromatic X-rays and propagation-based phase contrast to image hydrated tissues at sub-micron resolution.
- Deep Learning (CNN) Integration: Train self-supervised neural networks on KDE probability maps to accelerate 3D volumetric segmentation across noisy amber matrices.
- Multi-Species Taphonomic Modeling: Broaden the kinetic curing framework to soft-bodied taxa (e.g., arachnids, larvae) exhibiting variable cuticle permeability.
- Physical Calibration Phantoms: Pair scans with polymer and water references to improve calibration.

Summary of Contributions & Discussion

KEY DEFENSE TAKEAWAYS

Automated Segmentation

- Dynamic KDE Mode Suppression: Eliminates manual thresholding bias and discards the > 90% bulk resin attenuation peak slice-by-slice across synthetic and fossil matrices.

Contrast Resolution

- Multi-Energy Spectral PCA: Improves tissue contrast without chemical stains, delivering a 60.95 X quantitative information gain in PC_1.

Taphonomic Dynamics

- In-Situ Hydration Tracking: Proves internal negative-contrast air cavities develop during early resin contact (0-27h), confirming syn-depositional origin in fossil amber.

Acknowledgments

- Supervisor & Advisory: Academic guidance and methodology review.
- Imaging & Laboratory Facilities: Micro-CT instrument access and beam calibration support.
- Specimen Provision: Baltic amber fossil inclusion (Calliarcys antiquus sp. nov.) and reference materials.

Questions & Discussion

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